

MOHAMMED BAHAROON

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[Website](#) | [Scholar](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

Harvard University

Doctor of Philosophy, AI in Medicine Track, Biomedical Informatics

Boston, MA

2026 - Present

Harvard Medical School

Master of Medical Science, Biomedical Informatics

Boston, MA

2024 – 2026

Harvard Medical AI (HMAI) Group Founder and President

TF for BMIF203/BMI702: Foundations of AI in Medicine, Spring 2026

Penn State University

Bachelor of Science, Software Engineering

World Campus (Remote)

2021 – 2025

King Saud bin Abdulaziz University for Health Sciences

Bachelor of Health Informatics, Rank: 1/26

Riyadh, Saudi Arabia

2019 – 2023

Google DSC Founder and Lead, TEDx Technical Lead

PUBLICATIONS

- [1] **Baharoon, M.**, Heintz, T., Raissi, S., Alabbad, M., Alhammad, M., AlOmaish, H., ... & Rajpurkar, P. (2026). CRIMSON: A Clinically-Grounded LLM-Based Metric for Generative Radiology Report Evaluation. arXiv preprint arXiv:2603.06183. [Under review at MICCAI 2026](#).
- [2] **Baharoon, M.**, Raissi, S., Jun, J. S., Heintz, T., Alabbad, M., Alburkani, A., ... & Rajpurkar, P. (2025). RadGame: An AI-Powered Platform for Radiology Education. [Proceedings of the Machine Learning for Health \(ML4H\)](#). [Demo Oral Spotlight](#).
Also presented at ECR 2026 Oral; Harvard DBMI Science Day 2025; MIT-MGB AI Cures 2025.
- [3] **Baharoon, M.**, Luo, L., Moritz, M., Kumar, A., Kim, S. E., Zhang, X., ... & Rajpurkar, P. (2025). Rexgroundingct: A 3d chest ct dataset for segmentation of findings from free-text reports. arXiv preprint arXiv:2507.22030. [Under review at NEJM AI](#).
- [4] **Baharoon, M.**, Ma, J., Fang, C., Toma, A., & Wang, B. (2026). Exploring the Design Space of 3D MLLMs for CT Report Generation. [Medical Image Computing and Computer Assisted Intervention – MICCAI 2025](#) (pp. 237–246). Springer Nature Switzerland.
- [5] **Baharoon, M.**, Luo, L., Moritz, M., Kumar, A., Kim, S. E., Zhang, X., ... & Rajpurkar, P. (2025). State-of-the-Art Text-Prompted Medical Segmentation Models Struggle to Ground Chest CT Findings. In [Machine Learning for Healthcare Conference](#). PMLR.
- [6] **Baharoon, M.**, Klein, J., & Michels, D. (2025). Harmony: A Joint Self-Supervised and Weakly-Supervised Framework for Learning General Purpose Visual Representations. [Transactions on Machine Learning Research](#).
- [7] Ma, J., Yang, Z., Kim, S., Chen, B., **Baharoon, M.**, Fallahpour, A., ... & Wang, B. (2025). Medsam2: Segment anything in 3d medical images and videos. arXiv preprint arXiv:2504.03600. [Under review at Nature Biomedical Engineering](#).
- [8] Sambara, S., Kim, S. E., Zhang, X., Luo, L., Johri, S., **Baharoon, M.**, ... & Rajpurkar, P. (2025). 3DReasonKnee: Advancing Grounded Reasoning in Medical Vision Language Models. Pacific Symposium on Biocomputing.
- [9] Acosta, J., Adams, S., Kernbach, J., Hardy, R., Kim, S. E., Luo, L., Johri, S., **Baharoon, M.**, & Rajpurkar, P. (2025). Voice-guided Orchestrated Intelligence for Clinical Evaluation (VOICE): A Voice AI Agent System for Prehospital Stroke Assessment. International Conference on Artificial Intelligence for Medicine, Health, and Care.
- [10] Ma, J., Kim, S., Li, F., **Baharoon, M.**, Asakereh, R., Lyu, H., & Wang, B. (2024). Segment Anything in Medical Images and Videos: Benchmark and Deployment. <https://arxiv.org/abs/2405.14239/>.
- [11] **Baharoon, M.**, Almatar, H., Alduhayan, R., Aldebasi, T., Alahmadi, B., Bokhari, Y., ... & Aljouie, A. (2024). HyMNet: A multimodal deep learning system for hypertension prediction using fundus images and cardiometabolic risk factors. *Bioengineering*, 11(11), 1080.
- [12] **Baharoon, M.**, Qureshi, W., Ouyang, J., Xu, Y., Aljouie, A., & Peng, W. (2023). Towards General Purpose Vision Foundation Models for Medical Image Analysis: An Experimental Study of DINOv2 on Radiology Benchmarks. <https://arxiv.org/abs/2312.02366>.

RESEARCH EXPERIENCE

Harvard Medical School

Research Assistant - PI: Prof. Pranav Rajpurkar

Boston, MA

Feb 2025 – Present

- ★ Led a team of 15 radiologists and medical students to construct [ReXGroundingCT](#), the first dataset to link free-text findings to 3D chest CT segmentation, which is **under review at NEJM AI** and has been downloaded more than 3,000 times.
- ★ Adapted MedSAM2 for text prompts and benchmarked SoTA 3D segmentation models including SegVol, SAT, and MedSAM2 on ReXGroundingCT, revealing a large performance gap in grounding free-text findings; **published findings in MLHC 2025**.
- ◇ Created [RadGame](#), an AI-powered educational platform for radiology trainees. A user study of 18 medical students showed a 30–60% performance improvement across reporting and localization tasks; **published findings in ML4H 2025**.

Legend: ★ Chest CT Grounding ◇ AI-Powered Medical Education.

MIT Computer Science & Artificial Intelligence Laboratory (CSAIL)

Cambridge, MA

Research Assistant - PI: Prof. Regina Barzilay

Oct 2024 – Feb 2025

- Trained deep learning models for chest CT intra-subject deformable registration, to enable longitudinal tracking of lung nodule morphology.
- Designed a three-stage pipeline for lung nodule segmentation on NLST, integrating nodule detection, confidence-based classification, and final segmentation using the cropped detected region.

Vector Institute for Artificial Intelligence

Toronto, Canada

Research Intern - PI: Prof. Bo Wang

May 2024 – Aug 2024

- Proposed two novel radiology report enhancement methods that perform additional inferences using models trained to answer finding-specific questions, boosting CT report generation performance by over 10% on the GREEN score; **published findings in MICCAI 2025**.
- Achieved 2nd place in MICCAI's 2024 AMOS-MM competition for medical report generation and visual question answering using CT scans.
- Co-developed a semi-automatic ground-truth generation pipeline using MedSAM2 and evaluated the model across different modalities (CT, MRI, Echocardiogram); co-authored the [MedSAM2 paper](#) which is under review at **Nature Biomedical Engineering**.

King Abdullah University of Science and Technology (KAUST)

Thuwal, Saudi Arabia

Research Intern - PI: Prof. Dominik L. Michels

December 2023 – May 2024

- Developed [Harmony](#), a multi-objective weakly-supervised and semi-supervised learning paradigm that learns general purpose visual representations across tasks, outperforming CLIP, MaskCLIP, DetailCLIP, SLIP, and SILC; **published in TMLR**.
- Introduced a CLIP soft-target self-distillation method within Harmony that replaces the identity-matrix supervision with learned target distributions, addressing the one-to-one correspondence limitation of CLIP.

Fatima Fellowship

Remote

Research Fellow - Mentor: Dr. Wei Peng

Jun 2023 – December 2023

- Evaluated the robustness and generalizability of DINOv2 in the medical domain, conducting over 200 experiments across diverse modalities (X-ray, CT, and MRI), tasks (classification, segmentation), and in low and high data settings; shared findings in a [pre-print](#).

Tahakom

Riyadh, Saudi Arabia

Software Developer Intern

Mar 2023 – Jun 2023

- Developed a tailgating detection system using YOLOv7 with a +95% precision rate, tested throughout multiple days on live camera recordings using a NVIDIA Jetson edge device.

King Abdullah International Medical Research Center

Riyadh, Saudi Arabia

Research Intern - PI: Prof. Abdulrhman Aljouie

Oct 2022 – Mar 2023

- Devised [HyMNet](#): a multimodal deep learning systems that combine fundus photographs with demographic information to predict hypertension; **published findings in Bioengineering**.

AWARDS, HONORS, AND ACHIEVEMENTS

- Custodian of the Two Holy Mosques Scholarship, Ministry of Education, 160k USD 2024
- Nominated for Prince Faisal bin Bandar Award for Excellence and Creativity 2023
- [Google CSRMP Scholar](#), Google Research Class 2022B
- [The President Walker Award](#), Penn State University 2022

TEACHING AND TALKS

- Teaching Fellow at Harvard Medical School for the BMIF203/BMI702: Foundations of AI in Medicine graduate course, supervising projects related to computer vision and medical imaging. *Boston, Spring 2026*
- Featured as a guest on the 2nd episode of the Pixels 2 Patients podcast, discussing my work on ReXGroundingCT and RadGame. *Remote, 2025*

- Created educational videos reaching over 30,000 views on YouTube, [@Al-Jawhar](#). *Remote, 2023–Present*
- Taught a 4-day course titled "[Introduction to Programming with Python](#)" to 35 students at King Saud bin Abdulaziz University for Health Sciences. *Riyadh, 2023*
- Led a 2-day introductory course on machine learning for 20 students at King Saud bin Abdulaziz University for Health Sciences. *Riyadh, 2023*
- Presented a lecture on deep learning applications to over 400 participants during an online Artificial Intelligence Bootcamp. *Remote, 2022*
- Delivered a talk on computer vision with TensorFlow to over 200 attendees via Zoom as part of Google Developer Student Club events. *Remote, 2022*

ACADEMIC SERVICE

Reviewer

- *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* *2026*
- *Medical Image Analysis* (Elsevier) *2025*
- *Expert Systems with Applications* (Elsevier) *2025*
- *Machine Learning for Health (ML4H) Conference* *2025*

SKILLS

Programming: Python, PyTorch, C++, JavaScript, React, Django, PHP, HTML, CSS.

Research: Deep Learning, Computer Vision, Multimodal Machine Learning, Self-Supervised Learning, Medical AI.

Technologies: Linux, Git, Docker, Edge Computing (NVIDIA Jetson).